STAGE 1 — Install tools

```
Last login: Tue Nov 10 10:10:23 on ttys000
varnikakomali@VARNIKAs-MacBook-Air ~ % docker --version
zsh: command not found: docker
varnikakomali@VARNIKAs-MacBook-Air ~ % docker rm -f netlab
zsh: command not found: docker
varnikakomali@VARNIKAs-MacBook-Air ~ % cd desktop
varnikakomali@VARNIKAs-MacBook-Air desktop % docker rm -f netlab
zsh: command not found: docker
varnikakomali@VARNIKAs-MacBook-Air desktop % sudo ln -sf /Applications/Docker.app/Contents/Resources/bin/docker /usr/local/bin/docker
Password:
varnikakomali@VARNIKAs-MacBook-Air desktop % docker --version
Docker version 29.9.2, build a82e528
varnikakomali@VARNIKAs-MacBook-Air desktop % docker rm -f netlab
Docker run -it --privileged --name netlab ubuntu:latest bash
netlab
root@7e8c79fb6de8:/# apt update && apt install -y tshark iperf3 mtr-tiny iproute2 netcat-openbsd iputils-ping xxd
Get:1 http://ports.ubuntu.com/ubuntu-ports resolute InRelease [136 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports resolute-updates InRelease [137 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports resolute-backports InRelease [137 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports resolute-security InRelease [137 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports resolute/restricted arm64 Packages [231 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports resolute/main arm64 Packages [1860 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports resolute/multiverse arm64 Packages [298 kB]
Get:8 http://ports.ubuntu.com/ubuntu-ports resolute/universe arm64 Packages [19.9 MB]
Get:9 http://ports.ubuntu.com/ubuntu-ports resolute-updates/universe arm64 Packages [471 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports resolute-updates/restricted arm64 Packages [989 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports resolute-updates/main arm64 Packages [1118 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports resolute-updates/multiverse arm64 Packages [26.3 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports resolute-backports/universe arm64 Packages [3309 B]
Get:14 http://ports.ubuntu.com/ubuntu-ports resolute-security/multiverse arm64 Packages [24.5 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports resolute-security/main arm64 Packages [863 kB]
Get:16 http://ports.ubuntu.com/ubuntu-ports resolute-security/universe arm64 Packages [261 kB]
Get:17 http://ports.ubuntu.com/ubuntu-ports resolute-security/restricted arm64 Packages [893 kB]
Fetched 27.5 MB in 1min 26s (320 kB/s)
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  iperf3  iproute2  iputils-ping  mtr-tiny  netcat-openbsd  tshark  xxd

Installing dependencies:
  adduser                 libbcg729-0             libffi8                 libidn2-0               libldap2                libopencore-amrnb0      libsasl2-modules        libtbb12                libx11-6                publicsuffix
  ca-certificates         libbpf1                 libglib2.0-0t64         libiperf0               liblerc4                libopus0                libsasl2-modules-db     libtbbbind-2-5          libx11-data             shared-mime-info
  dbus                    libbrotli1              libglib2.0-data         libjansson4             liblua5.4-0             libp11-kit0             libsbc1                 libtbbmalloc2           libxau6                 wireshark-common
  dbus-bin                libcap2                 libgnutls30t64          libjbig0                libmaxminddb0           libpam-cap              libsctp1                libtiff6                libxcb1                 xdg-user-dirs
  dbus-daemon             libcap2-bin             libgpm2                 libjpeg-turbo8          libmnl0                 libpcap0.8t64           libsharpyuv0            libtirpc-common         libxdmcp6
  dbus-session-bus-common libcares2               libgrpc++1.51t64        libjq1                  libncurses6             libpciaccess0           libsmi2t64              libtirpc3t64            libxext6
  dbus-system-bus-common  libcurl4t64             libgrpc29t64            libjsoncpp26            libnettle8t64           libprotobuf32t64        libsnappy1v5            libunistring5           libxml2-16
  ibverbs-providers       libdbus-1-3             libgssapi-krb5-2        libk5crypto3            libnghttp2-14           libprotoc32t64          libspandsp2t64          libwebp7                libxnvctrl0
  krb5-locales            libdeflate0             libhogweed6t64          libkeyutils1            libnghttp3-9            libpsl5t64              libspeexdsp1            libwireshark-data       libxtables12
  libabsl20260107         libelf1t64              libhwloc-plugins        libkrb5-3               libnl-3-200             libre2-11               libssh-4                libwireshark19          linux-sysctl-defaults
  libapparmor1            libexpat1               libhwloc15              libkrb5support0         libnl-genl-3-200        librtmp1                libssh2-1t64            libwiretap16            ocl-icd-libopencl1
  libatomic1              libfalcosecurity0t64    libibverbs1             libldap-common          libnl-route-3-200       libsasl2-2              libtasn1-6              libwsutil17             openssl

Suggested packages:
  liblocale-gettext-perl  ecryptfs-utils          low-memory-monitor  krb5-user           libsasl2-modules-gssapi-mit  libsasl2-modules-sql  geoip-database              wireshark-doc
  perl                    default-dbus-session-bus  gnutls-bin        mmdb-bin            | libsasl2-modules-gssapi-heimdal  lksctp-tools  geoip-database-extra  opencl-icd
  cron                    | dbus-session-bus      gpm               opus-tools          libsasl2-modules-ldap        snmp-mibs-downloader  libjs-leaflet
  quota                   python3:any             krb5-doc          pciutils            libsasl2-modules-otp         geoipupdate           libjs-leaflet.markercluster
```

STAGE 2 — Build the network

```
root@7e8c79fb6de8:/# ip netns list
router (id: 2)
host1 (id: 1)
host2 (id: 3)
root@7e8c79fb6de8:/#
```

STAGE 3 — Configure IP Addresses

```
debconf: falling back to frontend: Teletype
Setting up libpsl5t64:arm64 (0.21.2-1.1build2) ...
Setting up libgrpc++1.51t64:arm64 (1.51.1-8ubuntu1) ...
Setting up libtbb12:arm64 (2022.3.0-2) ...
Setting up libhwloc-plugins:arm64 (2.13.0-2) ...
Setting up libirtmp1:arm64 (2.4+20151223.gitfa8646d.1-3) ...
Setting up libspandsp2t64:arm64 (0.0.6+dfsg-2.2build2) ...
Setting up libwsutil17:arm64 (4.6.4-1) ...
Setting up libpcap0.8t64:arm64 (1.10.6-1ubuntu1.1) ...
Setting up libcurl4t64:arm64 (8.18.0-1ubuntu2.8) ...
Setting up libwiretap16:arm64 (4.6.4-1) ...
Setting up libfalcosecurity0t64:arm64 (0.20.0-4ubuntu0.1) ...
Setting up libwireshark19:arm64 (4.6.4-1) ...
Setting up wireshark-common (4.6.4-1) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/aarch64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1
/usr/lib/aarch64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/aarch64-linux-gnu/perl-base /usr/lib/aarch64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/Fro
ntEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Setting up tshark (4.6.4-1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
Processing triggers for procps (2:4.0.4-9ubuntu1) ...
procps: Applying updated sysctl configuration
root@7e8c79fb6de8:/# ip netns add host1
ip netns add router
ip netns add host2

ip link add veth-h1 type veth peer name veth-r1
ip link add veth-r2 type veth peer name veth-h2

ip link set veth-h1 netns host1
ip link set veth-r1 netns router
ip link set veth-r2 netns router
ip link set veth-h2 netns host2
root@7e8c79fb6de8:/# ip netns list
router (id: 2)
host1 (id: 1)
host2 (id: 3)
root@7e8c79fb6de8:/# ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
ip netns exec host1 ip link set veth-h1 up
ip netns exec host1 ip link set lo up
root@7e8c79fb6de8:/# ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
ip netns exec router ip link set veth-r1 up
ip netns exec router ip link set veth-r2 up
ip netns exec router ip link set lo up
root@7e8c79fb6de8:/# ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
ip netns exec host2 ip link set veth-h2 up
ip netns exec host2 ip link set lo up
root@7e8c79fb6de8:/#
```

STAGE 4 — Enable Routing

```
root@7e8c79fb6de8:/# ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
ip netns exec host1 ip link set veth-h1 up
ip netns exec host1 ip link set lo up
root@7e8c79fb6de8:/# ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
ip netns exec router ip link set veth-r1 up
ip netns exec router ip link set veth-r2 up
ip netns exec router ip link set lo up
root@7e8c79fb6de8:/# ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
ip netns exec host2 ip link set veth-h2 up
ip netns exec host2 ip link set lo up
root@7e8c79fb6de8:/# ip netns exec router sysctl -w net.ipv4.ip_forward=1
ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
root@7e8c79fb6de8:/#
```

```
root@7e8c79fb6de8:/# ip netns exec router sysctl -w net.ipv4.ip_forward=1
ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
root@7e8c79fb6de8:/#
```

STAGE 5 — Test Connectivity

```
root@7e8c79fb6de8:/# ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.168 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.204 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.222 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2047ms
rtt min/avg/max/mdev = 0.168/0.198/0.222/0.022 ms
root@7e8c79fb6de8:/#
```

```
root@7e8c79fb6de8:/# ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.092 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.193 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.151 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2083ms
rtt min/avg/max/mdev = 0.092/0.145/0.193/0.041 ms
root@7e8c79fb6de8:/#
```

STAGE 6 — Add Delay and Loss

```
root@7e8c79fb6de8:/# ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1% seed 7854923334155970949
root@7e8c79fb6de8:/#
```

STAGE 7 & 8 — Start Simultaneous Captures

Termial1:

```
root@7e8c79fb6de8:/# ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1% seed 7854923334155970949
root@7e8c79fb6de8:/# docker exec -it netlab bash
bash: docker: command not found
root@7e8c79fb6de8:/# docker exec -it netlab bash
bash: docker: command not found
root@7e8c79fb6de8:/# docker exec -it netlab bash
bash: docker: command not found
root@7e8c79fb6de8:/# ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

Termial2:

```
Last login: Tue Nov 10 10:40:53 on ttys002
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
1
```

Terminal3:

```
Last login: Tue Nov 10 10:40:55 on ttys005
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

Terminal4:

```
Last login: Tue Nov 10 10:40:54 on ttys003
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec host2 nc -l 1234
```

Terminal5:

```
Last login: Tue Nov 10 10:40:54 on ttys003
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec host2 nc -l 1234
Hello from host1!
Three Point Capture Lab
```

STAGE 9 — Send real message from host1 to host2

```
Last login: Tue Nov 10 10:40:54 on ttys004
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec host1 nc 10.0.2.2 1234
Hello from host1!
Three Point Capture Lab
```

```
Last login: Tue Nov 10 10:40:54 on ttys003
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec host2 nc -l 1234
Hello from host1!
Three Point Capture Lab
```

STAGE 10 — What you see in real time

```
Last login: Tue Nov 10 10:40:54 on ttys003
varnikakomali@VARNIKAs-MacBook-Air desktop % /Applications/Docker.app/Contents/Resources/bin/docker exec -it netlab bash
root@7e8c79fb6de8:/# ip netns exec host2 nc -l 1234
Hello from host1!
Three Point Capture Lab
```

STAGE 12 — Fix permissions

```
root@7e8c79fb6de8:/# chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
root@7e8c79fb6de8:/# ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.2K Nov 10 05:19 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.1K Nov 10 05:19 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.0K Nov 10 05:19 /tmp/cap3.pcap
root@7e8c79fb6de8:/#
```

STAGE 13 — Check TTL

```
root@7e8c79fb6de8:/# chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
root@7e8c79fb6de8:/# ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.2K Nov 10 05:19 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.1K Nov 10 05:19 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.0K Nov 10 05:19 /tmp/cap3.pcap
root@7e8c79fb6de8:/# tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      4 63
      6 64
      4 63
      6 64
      6 63
      4 64
root@7e8c79fb6de8:/#
```

STAGE 14 — Read your actual message from the capture

```
root@7e8c79fb6de8:/# tshark -r /tmp/cap2.pcap -Y "tcp.port==1234" -T fields -e tcp.payload 2>/dev/null | xxd -r -p
Hello from host1!
Three Point Capture Lab
root@7e8c79fb6de8:/#
```

Delay checking

```
root@7e8c79fb6de8:/# tshark -r /tmp/cap2.pcap -Y "tcp.port==1234" -T fields -e tcp.payload 2>/dev/null | xxd -r -p
Hello from host1!
Three Point Capture Lab
root@7e8c79fb6de8:/# # Check relative arrival times of SYN packets
tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null

# Check retransmissions
tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l

# Check latency report
ip netns exec host1 mtr --report 10.0.2.2
66.018317294
0.000000000
0
0
0
Start: 2026-11-10T05:22:05+0000
```

Clean up:

```
root@7e8c79fb6de8:/# ip netns del host1
ip netns del router
ip netns del host2
ip netns list
root@7e8c79fb6de8:/#
```